

The empirical recurrence for OEIS A303244, recovered from its tail

Adrian Perez Fontelles

Independent researcher

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Abstract

OEIS A303244 records “Empirical recurrence of order 67” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 69$. Its last coefficient is $c_{67} = -28672 \neq 0$, so no further tail satisfies anything shorter; any order-67 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A303244 is “Number of $4 \times n$ 0..1 arrays with every element equal to 0, 2, 3, 5 or 6 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

8, 3, 55, 10, 255, 149, 1676, 3307, ...

The entry states:

Empirical recurrence of order 67 (see link above)

As of the “Last modified” line on the live entry (Apr 20 2018 10:57:15, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 67: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 67.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 2$ on, it returns order 67 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{67} c_i a(n-i),$$

c_1	5	c_{18}	23052609	c_{35}	−8771508460	c_{52}	−4027040687
c_2	27	c_{19}	−137773600	c_{36}	−3946468749	c_{53}	2182303177
c_3	−147	c_{20}	−60690230	c_{37}	9819586440	c_{54}	1887077245
c_4	−379	c_{21}	330195387	c_{38}	5974544061	c_{55}	−1217851920
c_5	2186	c_{22}	142379813	c_{39}	−9668732256	c_{56}	−678898680
c_6	3572	c_{23}	−695671724	c_{40}	−8604861964	c_{57}	531460244
c_7	−21526	c_{24}	−294123892	c_{41}	8258604039	c_{58}	178551136
c_8	−25033	c_{25}	1308361120	c_{42}	11279729048	c_{59}	−177277192
c_9	156203	c_{26}	531370711	c_{43}	−6168751739	c_{60}	−36110272
c_{10}	138040	c_{27}	−2234509285	c_{44}	−12928193771	c_{61}	43981072
c_{11}	−885072	c_{28}	−848556002	c_{45}	4380098869	c_{62}	5729120
c_{12}	−622474	c_{29}	3521010292	c_{46}	12523949827	c_{63}	−7390656
c_{13}	4061777	c_{30}	1243134710	c_{47}	−3575740660	c_{64}	−592896
c_{14}	2370392	c_{31}	−5164856097	c_{48}	−10155184496	c_{65}	711168
c_{15}	−15480089	c_{32}	−1775091172	c_{49}	3431066928	c_{66}	27136
c_{16}	−7844678	c_{33}	7033540847	c_{50}	6970860059	c_{67}	−28672
c_{17}	49881558	c_{34}	2603070046	c_{51}	−3059903881		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 69$, and it is the recurrence OEIS A303244 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{67} - c_1 x^{66} - \dots - c_{67}$. Its constant term is $-c_{67} = 28672$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 67 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 67, so $\deg q = 67$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 67, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 25 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 67, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -28672 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-67 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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